ModelIt!

Student Activity Pack

Why Can't Weather Apps Get It Right?

Modeling Atmospheric Systems, Chaos Theory, and the Limits of Prediction

Grade 10L1 | NGSS HS-ESS2-5, HS-ESS2-3 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think weather forecasts become less accurate for days further in the future?

2 What information do you think meteorologists use to predict tomorrow's weather?

3 Draw a diagram showing how you think a thunderstorm forms — what ingredients are needed?

4 Have you ever heard of the 'butterfly effect'? What do you think it means for weather prediction?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Solar Radiation Input | Air Temperature | Moisture Content | Wind Speed | Storm Probability

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Sunny and Stable

Solar Radiation: Moderate | Air: Dry and stable

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Approaching Front

Solar Radiation: Strong | Moisture: High and rising

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Butterfly Effect

Two runs with 0.1% difference in Solar Radiation

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Pressure Gradient — The difference in atmospheric pressure between two locations, which directly drives wind speed and direction — steeper gradients produce stronger winds and more dynamic weather

Wind Shear — The change in wind speed or direction with altitude, which is critical for severe weather development — strong wind shear helps organize thunderstorms into supercells capable of producing tornadoes

Land-Ocean Contrast — The difference in heating rate between land and water surfaces, which creates sea breezes, monsoons, and coastal weather patterns that complicate forecasting

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If we have supercomputers, satellites, and thousands of weather stations, why is a 10-day forecast still barely better than a coin flip? Use evidence from your simulation to support your answers.

1 How does your model demonstrate that weather is a chaotic system where small changes produce large effects?

2 Why does the butterfly effect set a fundamental limit on forecast accuracy that better technology cannot overcome?

3 What does the model reveal about why severe weather forecasting is harder than fair-weather forecasting?

4 How could understanding chaos theory help people make better decisions about weather-dependent activities?

5 What are the limitations of modeling the atmosphere with only five components?

8 STEM Challenge: Design a Severe Weather Alert System

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YOUR ENGINEERING MISSION

Design a localized severe weather monitoring and alert system that maximizes warning time for thunderstorms and tornadoes while minimizing false alarms.

THE SCENARIO

A school district in an area prone to severe thunderstorms and occasional tornadoes needs a better warning system than just watching the news. The current system provides only 13 minutes of average tornado warning time, and false alarms have trained people to ignore alerts. Your team must design a monitoring system that provides earlier, more accurate warnings for the school and surrounding community.

Engineering Design Thinking

1. WHAT SENSORS AND DATA SOURCES WILL YOUR SYSTEM USE AND WHERE WILL THEY BE PLACED?
2. HOW WILL YOU DEFINE ALERT THRESHOLDS THAT MAXIMIZE WARNING TIME WITHOUT TRIGGERING TOO MANY FALSE ALARMS?
3. WHAT COMMUNICATION CHANNELS WILL YOU USE TO REACH STUDENTS, STAFF, AND THE COMMUNITY QUICKLY?
4. HOW WILL YOU TEST AND VALIDATE YOUR SYSTEM BEFORE RELYING ON IT FOR SAFETY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Meteorologists study atmospheric science and forecast weather using computational models, satellite data, and field observations. They work for the National Weather Service, media organizations, private weather companies, and aviation, earning $55,000–$120,000/year. Atmospheric Scientists who research climate modeling and chaos theory earn $70,000–$140,000/year.